INCH-POUND

MIL-D-12491E
10 March 1989
SUPERSEDING
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29 June 1978

MILITARY SPECIFICATION

DEGREASERS: SOLVENT, TANK IMMERSION-TYPE

This specification is approved for use by the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tank-immersion, cold solvent degreaser with a safety cover and circulating solution filter.

1.2 Classification. The classification of the tank-immersion degreasers shall be as follows. The type to be furnished shall be as specified (see 6.2.1).

Type I - Pneumatic solution agitation.
Type II - Hydraulic solution agitation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-T-60 - Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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Characteristics requirements for

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-4109	- Coupling Halves, Low Pressure, Air Hose, Quick-Disconnect.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment, Requirement 4.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packaging, and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US DEPARTMENT OF LABOR

29 CFR 1910	- Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 - Dimension of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators.

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N. W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection (see 6.4) in accordance with 4.3. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The degreaser shall be new and of the contractor's current design. The degreaser shall consist of a cabinet to serve as a solvent reservoir, with base or legs, and shall have a safety closing cover. The degreaser shall have a soak tank or carrier platform with removable grating, and be equipped with an electrically powered constant flow circulating pump with replaceable cartridge filter, with flexible metal hose and nozzle to provide agitation with solvents at room temperature.

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3.2.1 Measurement system. The US customary system of units (US) or the International System of Units (SI) may be used in the design and construction of the machine. In this specification, all measurement, dimensions, sizes, and capacities are given in the US customary system of units (US). These measurements may be converted to the SI through the use of the conversion factors and methods specified in ASTM E380.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practices, shall be suitable for the intended purpose in the design and fabrication of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Raw materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Fabrication. The end item of equipment shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the end item of equipment, to its maximum rating and capacity, without permanent distortion.

3.4.1 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, or other defects. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.4.2 Welding, brazing, and soldering. All welding, brazing, and soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles cracks, gouges, holes, insufficient materials used. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance or repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

3.4.4 Threads. All machined threads shall conform to FED-STD-H28 and the applicable detailed standards referenced therein.

3.4.5 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

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3.4.6 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The degreaser shall comply with the general safety and health requirements promulgated under 29CFR 1910 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 5.5).

3.4.7 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.4.8 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.5 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be located for complete drainage and accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.6 Motor. Unless otherwise specified (see 6.2.1), the motor shall conform to NEMA MG-1, as applicable to the type degreaser specified. The motor shall be equipped with overload protection and rated for continuous duty operations, and the machine shall operate from a 115 volt, 60 Hertz, single phase, alternating current source. Starting switches, or other devices capable of producing an electric arc during normal operations shall not be used. The motor shall be completely compartmented and isolated within the cabinet and vented to the outside of the cabinet, or located completely outside the cabinet.

3.6.1 Electric power supply cable. The electric power supply cable shall be not less than ten (10) feet in length and shall terminate with a three (3) prong grounding plug in accordance with ANSI C73.11. The three (3) prong plug shall be furnished with a three to two prong outlet adapter, with a ground lead. The electric power supply cable shall include sheathing/covering in accordance with NFPA 70 and 29 CFR 1910. The electric power supply cable shall have a voltage/potential and ampere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, of the degreaser in accordance with NFPA 70.

3.6.2 Switch. The motor shall be equipped with an "off-on" switch, located in the motor compartment and accessible from the outside of the cabinet within access of the operator. The switch shall have a current rating of not less than 1-1/2 times the full load current rating of the motor. The switch shall be clearly and permanently marked to indicate the "on" and "off" positions.

3.6.3 Wiring. All wiring shall meet the requirements of NFPA 70. All wiring and electrical components shall be located a minimum of 18 inches above the floor.

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3.6.3.1 Fungus resistance. Electrical and electronic components and electrical insulation not otherwise specified herein shall be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not resistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Filter. The degreaser shall be equipped with a replaceable cartridge type filter rated at 50 microns to clean the solution passing through the flush hose. The filter shall be located above the maximum level of the solution in a position where it can be serviced without draining the tank.

3.8 Accessories. Unless otherwise specified (see 6.2.1), each degreaser shall be furnished with the following accessories.

3.8.1 Parts basket. Unless otherwise specified (see 6.2.1), each degreaser shall be provided with one parts basket constructed of 1/2 inch mesh, expanded metal not less than 0.050 inch thick and shall be of sufficient strength to support at least 150 pounds. The ends of the parts basket shall be constructed of sheet metal not less than 0.050 inch thick and shall be provided with a lifting handle on each end extending above the maximum level of the solution. The parts basket shall be designed to fit in the soak tank.

3.8.2 Drain shelf. A drain shelf that fits over the top edge of the cabinet end to give external drainage area for the baskets and parts shall be furnished. Solution drainage shall be back into the degreaser. The drain shelf shall permit unobstructed closing of the cover when the shelf is in place and the parts baskets resting thereon. The drain shelf shall be capable of supporting at least 150 pounds.

3.8.3 Brush. One fiber- or nylon-bristle, parts-cleaning brush shall be furnished along with a rack or holder to hold the brush when not in use.

3.9 Nameplate. The nameplate and information plate(s) can be incorporated into one or more corrosion-resistant metallic plate(s) at the machine manufacturer's option in accordance with MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured to the machine with permanent fasteners such as screws, bolts, and rivets. The nameplate shall be secured with permanent fasteners at each of the four corners. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number
Power input (volts, ampere, phase, Hertz, alternating current (AC)
Horsepower

3.10 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting.

3.11 Type I degreaser.

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3.11.1 Type I degreaser pneumatic solution agitation. The Type I degreaser shall have a capacity of not less than 20 gallons and not more than 40 gallons of solution. A soak tank shall be provided and equipped with an air manifold and connections for dispersing compressed air. Provision shall be made to allow overflow of solution and sludge to the main reservoir of the cabinet. The degreaser shall be provided with a quick disconnect air hose coupling conforming to MIL-C-4109, type as applicable, Class A, Style 1, size 1/4 NPT and a connection conforming to type as applicable Class A, Style 1, size 1/4 NPT for direct connection to an external air supply pressure of no less than 35 pounds per inch (PSI). An air control valve shall be provided to direct air through the soak tank air agitation device.

3.11.2 Pump. The pump for the Type I degreaser shall have a direct drive electric motor and shall be designed to handle fluids containing abrasive particles. If located inside the tank, the bearings and seals shall be located above the maximum level of the solution. The pump at the flush hose shall have a capacity of not less than 4 gallons per minute (gpm) delivered at the hose nozzle.

3.11.3 Base. The base shall be equipped with four antifriction-bearing, swivel-type casters which are impervious to oil and grease. The base and casters shall support the fully loaded degreaser when stationary or moving.

3.11.3.1 Load and shock. The degreaser while filled with solution and loaded with at least 150 pounds on each work shelf, shall withstand being traversed at a speed of at least 25 feet per minute over two obstacles at an angle such that only one caster at a time encounters each obstacle. The two obstacles shall consist of a vertical 1 inch drop and a 15 degree slope rising to a height of 1 inch (see Figure 1). The degreaser shall show no evidence of malfunction, leakage, permanent deformation or other damage after traversing the obstacles.

3.11.4 Work shelves. One or more work shelves shall be located above the maximum level of the solution but deep enough to minimize splashing. The shelves shall be removable or shall fold flat against the soak tank sides to give unobstructed access to the soak tank. The shelves shall each support 150 pounds without deformation.

3.11.5 Indicator light. The degreaser indicator light that denotes the electrical power is on shall be provided and be of the glow-plug type. The indicator light shall comply with the requirments of Underwriters Laboratories (UL).

3.11.6 Barrier filter. A barrier filter shall be provided. The barrier filter shall be rated at 150 micron.

3.11.7 Cover. The degreaser cover shall be of the fire-stop type. The cover shall be hinged along the back edge to permit raising to a vertical position. When upright, it shall be retained by a latch equipped with an automatic release device. The automatic device shall be heat activated, releasing the cover and allowing it to close at a temperature between 130 and 160 degrees Fahrenheit (F). The cover shall be spring loaded or otherwise designed and constructed to assure that it will close when the latch is manually or automatically released.

3.11.8 Cabinet. The degreaser cabinet shall be of sheet steel leak free and act as a reservoir for the solution in addition to containing the soak tank, with removable sludge collection trays, work shelves, pump, flush hose, and controls.

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3.11.9 Flush hose. The degreaser flush hose shall be of flexible metal construction and shall contain no rubber or fabrics. The hose shall be sufficiently rigid to remain in any position placed by the operator. The hose shall be of sufficient length so that the flow of solvent will reach all sections of the tank. The nozzle shall deliver a stream of solution not less than 3/8 inch in diameter.

3.12 Type II degreaser.

3.12.1 Type II degreaser hydraulic solution agitation. The Type II degreaser hydraulic solution agitation shall have a capacity of not less than 40 gallons nor more than 80 gallons of solution. A separate and removable soak tank shall be provided and equipped with a manifold for agitation of the solution. Provision shall be made to allow overflow of solution and sludge to the main reservoir of the cabinet.

3.12.2 Pump. The pump for the Type II degreaser shall have a direct drive electric motor, and shall be designed to handle fluids with abrasive particles without damage. Bearings and seals shall be located above the maximum level of the solvent. The output for hydraulic solution agitation and flush hose in operation shall be a total capacity of not less than 60 gpm, and shall direct a flow of solution of not less than 6 gpm through the flush hose.

3.12.3 Control valve. The control valve shall direct the pump output of solution either through the agitation manifold only or through the agitation manifold and through the flush hose simultaneously and shall be clearly and permanently marked to indicate the operation position.

3.12.4 Base. The base shall be as specified in 3.11.3 and 3.11.3.1.

3.12.5 Work shelves. The work shelves shall be as specified in 3.11.4.

3.12.6 Indicator light. The indicator light shall be as specified in 3.11.5.

3.12.7 Barrier filter. The barrier filter shall be as specified in 3.11.6.

3.12.8 Cover. The cover shall be as specified in 3.11.7.

3.12.9 Cabinet. The cabinet shall be as specified in 3.11.8.

3.12.10 Flush hose. The flush hose shall be as specified in 3.11.9.

3.13 Workmanship. Standards of workmanship shall assure that the degreaser shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified in the contract, first article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all the tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using Inspection Levels and Acceptable Quality Levels (AQL), as listed, unless 100 percent inspection is specified.

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	II	1.0
Operational test (4.6.1)	100 percent	----
Automatic release device	100 percent	----
Packaging inspection (4.7)	S-4	4.0

4.5 Product examination. Visually, dimensionally, and manually examine each item to determine conformance with the requirements of 3.2 through 3.5, 3.8 through 3.10, and 3.13. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to identified specified standards, adequacy of markings, proper cleaning, and freedom from defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the item and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition,

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failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.7), shall be cause for rejection.

4.6. Tests.

4.6.1 Operation. Fill degreaser with solution to the operating level, start and run the pump for 30 minutes and operate degreaser while pump is operating as follows:

(1) Type I. Cut in air supply and agitate solution for 30 minutes.
(2) Type II. Direct pump output of solution through agitation manifold only for 15 minutes and through both manifold and flush hose simultaneously for 15 minutes.

Any one of the following defects shall constitute failure of this test:

(a) Leakage of solution in hydraulic system, soak tank, or cabinet seams.
(b) Inability to place flush hose in a stationary position or to reach any section of the soak tank. Types I and II only.
(c) Failure to have required agitation of the solution in hydraulic or mechanical Type II degreasers, or after supplying compressed air to pneumatic Type I degreasers.
(d) Failure of the pump to circulate solution at a rate of 60 gpm total output and not less than 4 gpm through the flush hose when valved into the flow pattern for Type II degreasers, and at a rate of not less than 4 gpm through the flush hose for Type I degreasers.

4.6.2 Automatic release device. Place the cover in the fully open position. Heat the automatic release device until the cover is released. Measure and record temperatures throughout the test. Release of the cover below 130 degrees F or failure of the cover to close at 160 degrees F shall constitute failure of this test (see 3.11.7).

4.6.3 Load and shock test. The Type I and Type II degreaser shall be filled with solution to operating level and each shelf loaded with 150 pounds of weight. The load and shock test shall be conducted in accordance with method specified in Figure 1. The degreaser shall be traversed across a 1-inch depression (obstacle No. 1) at a 15 degree angle and at a minimum rate of travel of 25 feet per minute. The movement shall be that one caster at a time traverses the depression. The degreaser shall then be moved up and over obstacle No. 2 at 15 degree slope 1 inch high in the same direction at a minimum travel of 25 feet per minute. Obstacles Nos. 1 and 2 shall be of hard wood, concrete or steel. Breakage or deformation of the casters or any part of the degreaser shall constitute failure of this test (see 3.11.3.1).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Level C as specified (see 6.2.1).

5.1.1 Level A.

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5.1.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Unpainted metal surfaces. Unless otherwise specified herein, unpainted exterior metal surfaces of the degreaser and accessories requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative.

5.1 1.3 Pump and accessories. Interior surfaces of the pump, filter, and discharge hose shall be coated with P-10, Type I, Grade 30 preservative. The pump shall be actuated to assure coating of all interior surfaces and parts.

5.1.1.4 Electric cable. The electric cable shall be coiled to the minimum diameter and the coil shall be secured with evenly spaced wraps of tape conforming to PPP-T-60, Type IV.

5.1.1.5 Casters and valves. Casters shall be removed and drain valves open.

5.1.2 Level C. Each complete degreaser shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be Level A, Level B, or Level C as specified (see 6.2.1).

5.2.1 Level A. Each complete degreaser, preserved as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Style A or B. Cushioning, blocking, and bracing shall be provided to prevent damage to the degreaser.

5.2.2 Level B. Each complete degreaser preserved as specified in 5.1 shall be packed as specified for Level A, except the box shall be Domestic Type, style optional.

5.2.3 Level C. Degreasers preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1190.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The degreasers are intended for use in removing grease and dirt from machinery, castings, and parts associated with maintenance shop operation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type degreaser required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and

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2.2).
(d) First article when required (3.1).
(e) Exceptions and additional safety/health requirements when required (see 3.4.6 and 6.5).
(f) Motor, if other than that specified (see 3.6).
(g) Accessories required (see 3.8).
(h) Parts baskets mesh size if other than as specified (see 3.8.1).
(i) Painting, if different (see 3.10).
(j) First article inspection, if different (see 4.3).
(k) Quality conformance inspection, if different (see 4.4).
(l) Level of preservation and packing, if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.6 Subject term (key word) listing.

Agitation
Pump
Cabinet
Solvent

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodians:
Army - AL
Navy - YD

Review activity:
Navy - SH
DLA - CS

Preparing activity:
Army - AL

Project No. 4940-0589

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drawings and to the ... to the limits specified herein, and on applicable

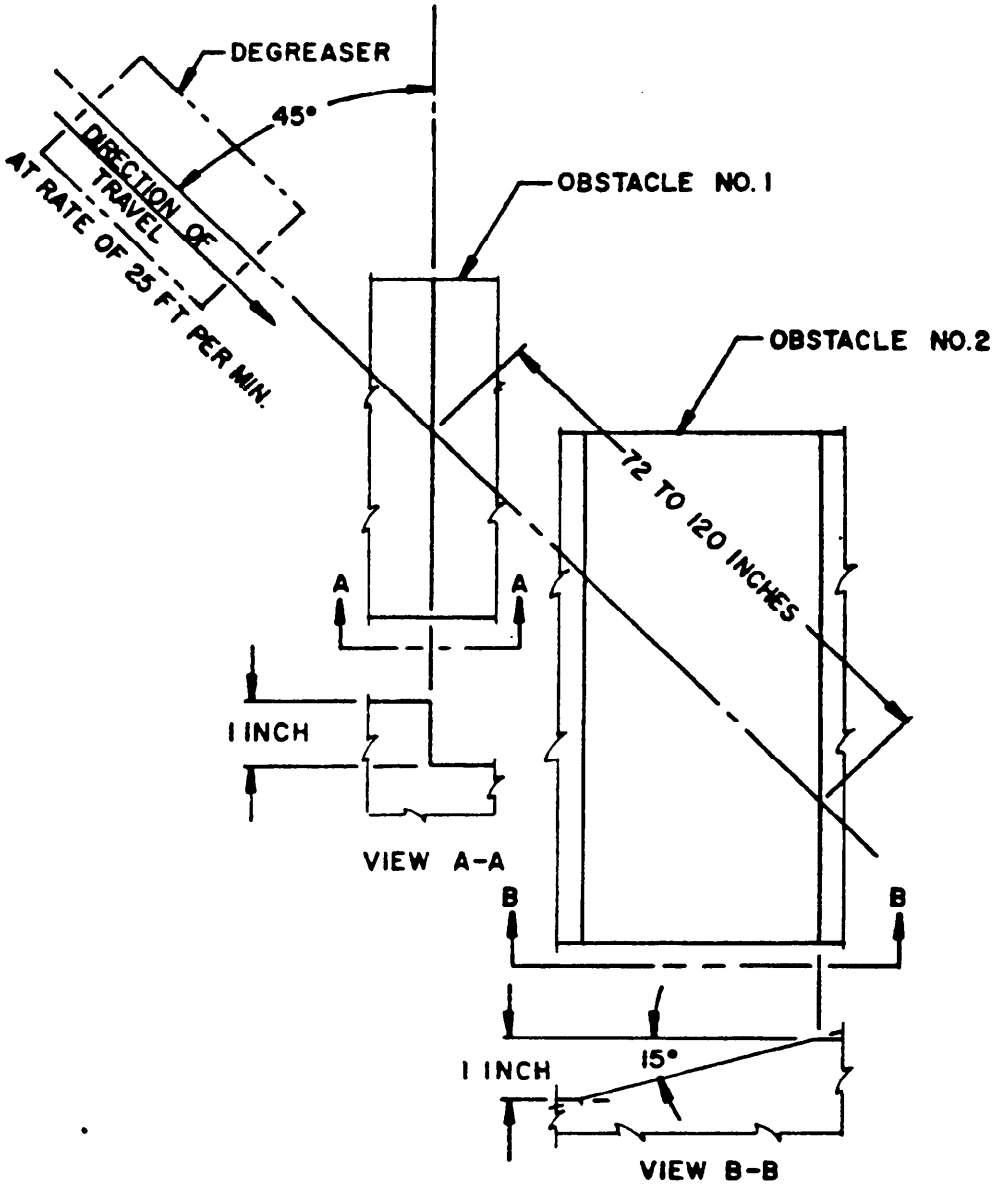


FIGURE 1. LOAD AND SHOCK TEST METHOD

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